

Duality

We can regard all arrows in a category as pointing the other way, and this gives us the dual category. One advantage is that we immediately get a dual version of every construction and every theorem.

17.1 Turning arrows around

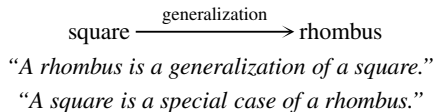
When we draw an arrow $\longrightarrow$ normal convention has it that we are indicating a direction, in this case moving towards the right. This may be something to do with the shape of actual arrows (as in the weapon) and that this is an abstract diagram of one of those, but I imagine that most of us have not seen very many arrow-weapons in real life. I've seen a few in museums but that's it.

This association of direction with the diagram is just a convention. We could take the convention to be the other way without changing anything about the structure of the situation, we'd just risk confusing everyone.

Duality in category theory is about doing just this: turning all the arrows around without really changing the structure of the situation. It also risks confusing everyone, but when the concepts in question are very abstract it's sometimes clear that the choice of direction is arbitrary.

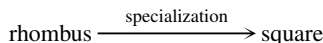
One example we saw early on, in Chapter 5, was the diagram of types of quadrilateral. We were thinking about generalizations and drew each arrow to represent relaxing a condition and thereby moving from a special case to a more general one.

Here is an example of one of those arrows, and two different ways we can read the same information.



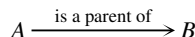
These two readings are not expressing any different information, they're just expressing it the other way round.

We could also draw it the other way round, like this.



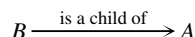
Things To Think About

T 17.1 Here is an example of a family relationship we saw in Chapter 5.



If we turn this arrow around what will the relationship be? Is it the opposite?

If we turn around the parent arrow we get this:



In both the generalization and parent examples it might seem that the two directions are opposites of each other, but opposites are a hazy concept. I suppose if we asked someone in normal life “What’s the opposite of a parent?” they would probably say a child, but we could also say it’s a non-parent. Duality, by contrast, is a precise concept.

Things To Think About

T 17.2 What relationships do we get if we turn these arrows round?

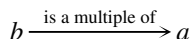
1. $a \xrightarrow{\text{is a factor of}} b$

3. $a \xrightarrow{\text{gain of one type of privilege}} b$

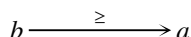
2. $a \xrightarrow{\leq} b$

4. $a \xrightarrow{\sim} b$ (indicating an isomorphism)

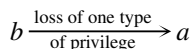
If we turn around the arrow for “is a factor of” we get:



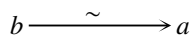
For $\leq$ we get $\geq$, which is not exactly the “opposite” as both versions include the $=$ part.



For the third example, gain of privilege becomes this:



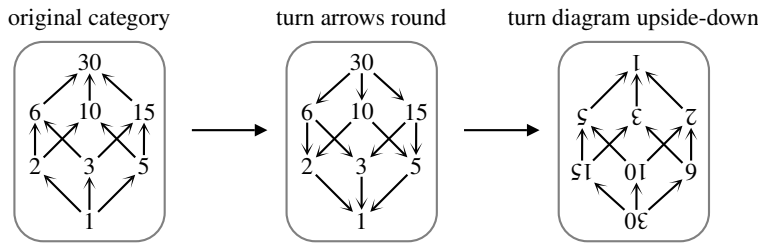
For the last example, we get the same relationship:



So far all we’ve done is turn around individual arrows, but if we then look at the effect on structures, more interesting things happen.

Things To Think About

T 17.3 Take a diagram of factors of 30, turn every arrow around, and then physically turn the new diagram upside-down. What do you notice if you compare this with the original diagram? Which numbers play analogous roles? What are the relationships between the pairs of numbers x, y where the role of x in the first diagram is analogous to the role of y in the last?



Probably the most immediately obvious thing is that the last diagram has the same shape as the first one — it’s a cube, so upside-down it’s still a cube. This means every object has an analogous object in the other diagram. 1 and 30 have switched places (they’re opposite corners of the cube), which means that the terminal object and initial object have switched places.

We then see that we have the following analogous numbers.

upper row		lower row
6	is analogous to	5
10	is analogous to	3
15	is analogous to	2

It turns out these two tables give us the same information, just stated a different way round. Moreover each pair of analogous numbers multiplies together to make 30. That is a more standard school-level way of thinking about factors: in pairs that multiply together to make the number in question.

All of these features are about duality and are a way we can glean more understanding from situations by looking at them the other way round.

17.2 Dual category

We have made many references to this idea of turning arrows around but now we are finally going to present it formally. The basic concept is that given any category, there is a related category where we simply consider every arrow to be facing the other way. The information isn’t different, just the presentation, but it’s a remarkably powerful concept. We use the notation $\mathcal{C}^{\text{op}}$, pronounced “C op”. The “op” stands for “opposite”.

Definition 17.1 For any category $\mathcal{C}$ the *dual (or opposite) category* $\mathcal{C}^{\text{op}}$ is given as follows.

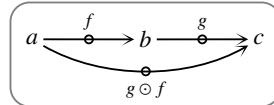
- objects: $\text{ob } \mathcal{C}^{\text{op}} = \text{ob } \mathcal{C}$
- morphisms: given objects a, b , we have $\mathcal{C}^{\text{op}}(a, b) = \mathcal{C}(b, a)$
- identities are the same
- composition is reversed (which I’ll explain below).

In order to explain the last point, I first want to give a little more intuition about what this definition means. This can all get very confusing so I'm going to draw arrows in $\mathcal{C}^{\text{op}}$ with a little circle on them, like this: $\text{---}\circ\text{---}$

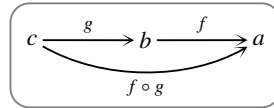
The definition says an arrow $a \xrightarrow{\text{---}\circ\text{---}} b \in \mathcal{C}^{\text{op}}$
 is given by an arrow $b \xrightarrow{\text{---}} a \in \mathcal{C}$

The “meaning” of the arrow is no different, just the direction we're regarding it as having, formally. Now for composition in $\mathcal{C}^{\text{op}}$ I am going to write $\odot$.

We need to consider this configuration in $\mathcal{C}^{\text{op}}$



The diagram this corresponds to in $\mathcal{C}$ is this:



So it makes sense for us to define composition in $\mathcal{C}$ by: $g \odot f = f \circ g$.
 This is what I mean by “composition is reversed” in the definition.

Note on type-checking

Here, “makes sense” means that the composite $f \circ g \in \mathcal{C}$ has the correct source and target to be a candidate for giving the composite $g \odot f \in \mathcal{C}^{\text{op}}$. This is called *type-checking*, which is a term that is more prevalent in computer science. When you ask a computer to handle a variable, you first have to tell it what type of variable to expect: is it going to be a number, a letter, a string, and so on. If you then ask the computer to perform a procedure that produces a variable that is not of the type the computer was expecting then the computer will get confused as your procedure “doesn't type-check”.

Category theory puts so much structure in place in advance that often a large part of a construction or proof is just a question of type-checking. Sometimes we're guided in our arguments or definitions because there's only one thing that will type-check, as in the above situation. We couldn't even try and make $g \odot f = g \circ f$ because it doesn't type-check: f and g are not composable that way round in $\mathcal{C}$, they are only composable the other way round. Type-checking can pre-empt a lot of muddy thinking. It's a bit like starting cooking by getting out all your ingredients, measuring them into little bowls, and lining them up neatly like they seem to on television cooking shows. I'm usually too lazy/impatient to do that but I have to admit that if I can be bothered to do it, it makes the actual cooking process more calm and organized.

So far I have written out all the arrows in $\mathcal{C}^{\text{op}}$ with circles on them, but in general I prefer to completely avoid drawing arrows in the dual category. Apart from in the definition itself, it's rarely necessary, and I usually find it more confusing than it's worth. Usually we use dual categories as either a thought process or a way of formalizing the idea of functors that flip the direction of arrows, as we'll see in Chapter 20.

We say “dually” to mean “now repeat this argument but in the dual category”. But rather than draw the arrows in the dual category, we flip the arrows in the argument, as we'll see in the following examples. The idea is that given any construction or argument that we have made in a category $\mathcal{C}$, there is a dual version which consists of doing the exact same thing but in the category $\mathcal{C}^{\text{op}}$. The dual version is often indicated by the prefix “co-”. So for example:

- we define products and then the dual concept is coproducts
- we define limits and then the dual concept is colimits
- we define equalizers and then the dual concept is coequalizers
- we define projections and then the dual concept is coprojections.

This gives rise to all sorts of silly jokes involving either sticking the prefix “co-” onto a word in normal English, or re-interpreting words that start with “co” in normal English but not as a prefix, such as “countable” (co-untable: what does “untable” mean?). Here are a couple of jokes. If you get them then you're getting the idea of duals.

1. What do you call someone who reads category theory papers? A co-author.
2. A mathematician is a machine for turning coffee into theorems. . .
 . . . A co-mathematician is a machine for turning co-theorems into ffee.

The last joke depends on the fact that the dual of a dual takes us back to the original concept.

Proposition 17.2 *Let $\mathcal{C}$ be any category. Then $(\mathcal{C}^{\text{op}})^{\text{op}} = \mathcal{C}$.*

I experienced a little shudder there because I wrote an *equality* between categories! This is two levels too strict: the correct notion of sameness between categories, which we aren't ready to define formally yet, is equivalence. Isomorphism is a little less nuanced, and equality is dire. And yet I think this really is an equality because it was defined via some equalities of sets, which is already one level stricter than morally correct.

17.3 Monic and epic

We informally pointed out that the definition of epic is just the same as monic with the arrows turned around, but here is what that means formally.

Proposition 17.3 *A morphism f is epic in $\mathcal{C}$ if and only if it is monic in $\mathcal{C}^{\text{op}}$.*

Maybe this should be the definition of epic, but given that we defined epic directly, this is duly a proposition. Also the word “it” needs some care: the morphism f is a morphism in both $\mathcal{C}$ and $\mathcal{C}^{\text{op}}$ because the morphisms in the dual category are the same as those in $\mathcal{C}$, just regarded the other way round.

This gives us a model for what “dually” means: it means we take the concept in question, that we have studied in $\mathcal{C}$, and now study it in $\mathcal{C}^{\text{op}}$, but still draw the arrows in $\mathcal{C}$. The following steps take us from the definition of monic to the dual definition of epic.

The definition of a monic in $\mathcal{C}$ involves this shape of diagram:

$$m \rightrightarrows a \longrightarrow b \in \mathcal{C}$$

So a monic in $\mathcal{C}^{\text{op}}$ involves a diagram like this:

$$m \begin{array}{c} \xrightarrow{s} \\ \xleftarrow{t} \end{array} a \xrightarrow{f} b \in \mathcal{C}^{\text{op}}$$

Now, following the principle of only drawing arrows in $\mathcal{C}$ and not in $\mathcal{C}^{\text{op}}$, we draw this as:

$$m \begin{array}{c} \xleftarrow{s} \\ \xrightarrow{t} \end{array} a \xleftarrow{f} b \in \mathcal{C}$$

Finally, as we prefer arrows to point right, we flip it physically on the page, giving the usual diagram for the definition of epic:

$$b \xrightarrow{f} a \begin{array}{c} \xrightarrow{s} \\ \xleftarrow{t} \end{array} m \in \mathcal{C}$$

Things To Think About

T 17.4 What is the dual of an isomorphism? That is, if f is an isomorphism in $\mathcal{C}^{\text{op}}$ what does that tell us about it in $\mathcal{C}$?

The diagram on the right shows f as an isomorphism, with inverse g . If we turn the arrows around it's the same diagram.

$$a \begin{array}{c} \xrightarrow{f} \\ \xleftarrow{g} \end{array} b$$

So if f is an isomorphism in $\mathcal{C}^{\text{op}}$ then it is also an isomorphism in $\mathcal{C}$. In fact that is exactly the same information. If a concept gives the same information whether it's in $\mathcal{C}$ or in $\mathcal{C}^{\text{op}}$ then it is called *self-dual*.

Another dual situation we saw in Chapter 15 on monics and epics was the proof that every isomorphism is monic: we rather briefly said the proof that every isomorphism is epic follows “dually”.

Things To Think About

T 17.5 Can you state this formally using the dual category?

The formal version of the dual proof goes like this. We have proved that if f is an isomorphism in $\mathcal{C}$ then it is monic in $\mathcal{C}$. But this is true in any category, so it is also true in $\mathcal{C}^{\text{op}}$. Now if f is an isomorphism in $\mathcal{C}$ then it is an isomorphism in $\mathcal{C}^{\text{op}}$, thus it is a monic in $\mathcal{C}^{\text{op}}$, but this precisely says that it is epic in $\mathcal{C}$.

This is an example of what I call a BOGOF[†] on theorems.

Duality BOGOF principle

Any concept in a general category can be considered, in particular, in $\mathcal{C}^{\text{op}}$, and then translated back into a concept in $\mathcal{C}$ with no extra proof or argument needed. This is true of definitions and proofs. We just say “and dually”.

Here’s an example of how we get to use the BOGOF principle.

Things To Think About

T 17.6 Suppose we have morphisms $a \xrightarrow{f} b \xrightarrow{g} c$ in $\mathcal{C}$.

1. Prove that if f and g are both monic then the composite gf is monic.
 2. What does the dual give us for free?
-

Proof To show that gf is monic we need to consider a diagram as on the right, and show that $gfs = gft \implies s = t$.

$$m \begin{array}{c} \xrightarrow{s} \\ \xrightarrow{t} \end{array} a \xrightarrow{f} b \xrightarrow{g} c$$

So suppose $gfs = gft$. Since g is monic we know that $fs = ft$. Since f is monic we know that $s = t$ as required.

The dual tells us that if f and g are both epic then gf is epic. □

Note that if we really translated this result into $\mathcal{C}^{\text{op}}$ we might get rather confused about the direction of the composite, which in $\mathcal{C}^{\text{op}}$ would be written fg . This is one of the reasons I prefer not to write down arrows in the dual category. Another reason (less utilitarian and more ideological) is that I find that the *point* of considering the dual is to think about new diagrams in $\mathcal{C}$, not pretend you’re in a different category temporarily.

Here’s another BOGOF example.

Things To Think About

T 17.7 Given morphisms $a \xrightarrow{f} b \xrightarrow{g} c$ in $\mathcal{C}$, prove that if gf is monic then f must be monic but g need not be. [Hint: the first part follows by working your way through the definitions. For the second part you could try finding a counterexample in **Set** using injective[‡] and non-injective functions.] What does the dual give us for free?

[†] Buy One Get One Free (also called BOGO in the US).

[‡] Reminder: injective means “no output is hit more than once”.

Proof Again we are considering the following diagram:

$$m \begin{array}{c} \xrightarrow{s} \\ \xrightarrow{t} \end{array} a \xrightarrow{f} b \xrightarrow{g} c$$

Suppose that gf is monic. To show that f is monic we suppose $fs = ft$. But this certainly means $gfs = gft$ and then as gf is monic we can deduce that $s = t$ as required.

To show that g need not be monic let us consider this in **Set** and find functions f and g such that gf is injective but g is not.

The idea is that f has to be injective because if it mapped two different things to the same place then gf would be doomed to do the same, whereas g doesn't need to be injective on the non-image[†] of f , because the non-image of f never gets in on the action of gf . So we just need b to be big enough to have some non-image of f .

Consider the sets and functions shown on the right. The composite gf is monic (in fact it's an isomorphism) but g is not, so this is a counterexample. Dually, if gf is epic then g must be epic but f need not be. $\square$

$$\begin{array}{ccccc} a & \xrightarrow{f} & b & \xrightarrow{g} & c \\ 0 & \dashrightarrow & 0 & \dashrightarrow & 0 \\ & & 1 & \nearrow & \end{array}$$

The BOGOF dual part perhaps requires some care because of the directions.

Here is a line-by-line comparison of the argument in $\mathcal{C}^{\text{op}}$ and how it translates to $\mathcal{C}$.

$\in \mathcal{C}^{\text{op}}$	$\in \mathcal{C}$
$a \xrightarrow{f} b \xrightarrow{g} c$	$a \xleftarrow{f} b \xleftarrow{g} c$
if the composite is monic then f must be monic	if the composite is epic then f must be epic

I find that the best way to avoid confusion is to detach my brain from the letter names of the functions and think of their position instead: if a composite is monic then the *first* arrow you travel along must be monic, and if a composite is epic then the *last* arrow you travel along must be epic.

Things To Think About

T 17.8 Although it's not logically necessary, can you find a counterexample in **Set** to exhibit the fact that gf can be epic even if f isn't epic? Look back at the example for monics.

The example we gave for monics also works for epics. The composite gf was epic (it was an isomorphism) but f was not epic.

These proofs involving monics, epics and composites are things I really enjoy again and again no matter how many times I do them. It's not for their

[†] Reminder: the image is "all the outputs that are actually hit", and non-image means "outputs that are not hit".

usefulness, it's not particularly because of what they illuminate, it's not because of profound applications or relevance to my life. I just find them deeply satisfying. It's the sense of fitting logical pieces together perfectly. This is why I loved category theory on impact, the very first time I studied it. It was the purest logical joy I had encountered. Loving it for its illuminating and applicable aspects came much later.

17.4 Terminal and initial

Another situation in which we thought about turning arrows around was when we started with the definition of initial object and then “turned the arrows around” to make the definition of terminal object.

Things To Think About

T 17.9 Can you make that correspondence between terminal and initial objects precise using the formal concept of $\mathcal{C}^{\text{op}}$?

Proposition 17.4 *Let $\mathcal{C}$ be a category. A terminal object in $\mathcal{C}$ is precisely an initial object in $\mathcal{C}^{\text{op}}$.*

Note that as with epics, I might prefer this to be the *definition* of terminal object, but as we already defined them directly this is a result that follows. It's another of those situations where we can choose which is the definition and which is the result. Anyway, this is what we mean by saying that terminal objects are the dual of initial objects. Indeed some people call them final objects and cofinal objects.

Every universal property we look at (well, also all the ones we don't look at) will come with a dual version, and from now on we will typically say that first, and then unpack what the dual means when translated back into $\mathcal{C}$.

17.5 An alternative definition of categories

I want to finish this chapter by pointing out where this duality in categories really comes from, or rather, by presenting a slightly different point of view on the definition of category that makes the concept of duality jump out at us. There are many different ways of presenting the underlying data for a category, emphasizing different aspects. Depending on how you present the underlying data, you then have to present the structure and the properties slightly differently too, but I want to focus on the data for now.

We previously defined a (small) category $\mathbb{C}$ to have

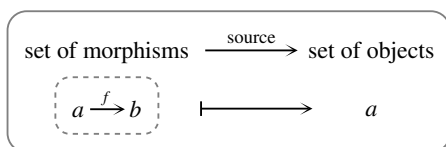
- a set of objects, and
- for every pair of objects a, b a set of morphisms $\mathbb{C}(a, b)$.

However, we never really talked about the “set of all morphisms”, just the set of morphisms from any given object to another. Instead we could say there is

- a set of objects C_0 , and
- a set of morphisms C_1

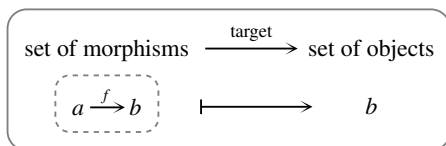
and now we need to do a bit of extra work to say that each morphism goes from one object to another. We express this using functions.

The idea is that every morphism has a source object, so we can make a function as shown on the right.

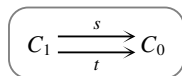


The small dotted box is to indicate that the morphism f is one element in the set of “inputs” here, as the set of inputs is the set of morphisms. It is mapped to the “output” a .

Likewise we have a function sending each morphism to its target object, as shown here.



So, going back to the notation of the sets C_0 and C_1 we have two functions like this.



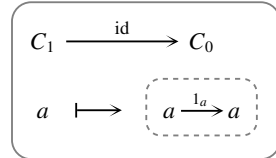
I hope it is now clear that there is symmetry between s and t . Each one is merely a function from C_1 to C_0 , and flipping the morphisms of the category $\mathbb{C}$ just consists of swapping the names s and t here.

This different way of describing the underlying data for a category is particularly important for certain kinds of generalization, including into higher dimensions. This diagram of two sets and two functions between them is sometimes called a graph or a directed graph, not to be confused with the sorts of graph you are made to sketch at great length at school. The idea here is that we could draw out the information contained in this “graph” as a diagram of actual arrows pointing from and to their designated sources and targets. It’s called directed because the source and target functions give direction to the arrows. In category theory we rarely consider undirected graphs, so we’re more likely to call this a graph and leave it to be specified when it’s *not* directed.

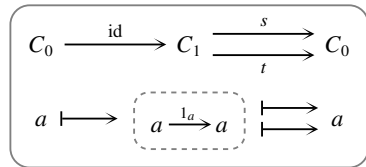
Things To Think About

T 17.10 The next step in turning this into a full definition of category would be to work out how to express identities and composition. Try making a function that takes an object a and produces the identity on it as output. What conditions must this function satisfy to ensure that the identity 1_a has the correct source and target?

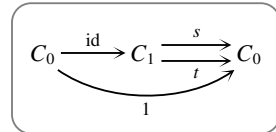
To express identities we need a function as shown here, with “id” for identity.



In order to be sure that the identity on a really has a as its source and target, we need to apply the source and target functions after the identity function, and know that we get a back again.



We can express this condition by using the diagram on the right, and asking that the composite functions $s \circ \text{id}$ and $t \circ \text{id}$ are both the identity function 1 (which sends every element to itself).



We’re not ready to develop this into a full definition of categories yet as we need to look at another universal property first, in order to define composition. We’ll see that in the next chapter.